Question 8

Not yet answered

Marked out of 1.00

Flag question

Transfers of data in I/O operation can happen synchronously on _____ and asynchronously on _____.

- data bus, address bus
- I/O bus, memory bus
- address bus, data bus
- memory bus, I/O bus

Question 10

Not yet answered

Marked out of 1.00

Flag question

What is the capacity of the main memory, if the total number of blocks in the memory is 197? Consider each blocks consist of 6 words of 1 byte size each.

Select one:

- 9.456 Kbits
- 1182 Bits
- 1.576 Kbits
- 197 KBits

Question 11

Not yet answered

Marked out of 2.00

Flag question

System A is normalized to System B to obtain the average performance. Consider the execution time of programs P1, P2, and P3 on system A, which are normalized to systems B were 0.25 seconds, 0.67 seconds, and 1.00 seconds, respectively. Calculate the geometric mean of system A.

- 1.24
- 1.39
- 1.92
- 0.64

Question 13

Not yet answered

Marked out of 1.00

Flag question

Operations performed by a CPU such as executing the instruction cycles, performing an arithmetic operation, and so on, are governed by a _____.

- input devices
- Control Unit (CU)
- interconnection bus
- system clock

Question 12

Not yet answered

Marked out of 1.00

Flag question

The _____ is more helpful than the arithmetic average when comparing the relative performance of two computer systems.

- geometric mean
- instruction count
- discrete time
- synthetic benchmark

Question 14

Not yet answered

Marked out of 1.00

Flag question

Refer to the following Fetch cycle, choose a micro-operation that will not occur at t_2.

t_1: $MAR \leftarrow IP/PC$

t_2: $MBR \leftarrow [MAR];$

$IP/PC \leftarrow IP/PC + 1$

t_3: $IR \leftarrow MBR$

- The program counter is incremented by one
- Content from program counter is loaded into MBR
- Instruction is accessed and transferred to the CPU through data bus
- The instruction address is loaded into MBR

Time left 0:16:02

Question 16

Not yet answered

Marked out of 1.00

Flag question

Which of the following is INCORRECT about Random Access Memory (RAM)?

Select one:

- RAM is volatile, and loses this information once the power is turned off
- RAM is used to store program during execution
- RAM also known as primary memory
- RAM is used to stores critical information

Question 15

Not yet answered

Marked out of 1.00

Flag question

_____ is the most common approach to assessing the processor and improvement of computer system performance.

- Geometric mean
- Statistical tools
- Benchmarking
- Arithmethic mean

Question 17

Not yet answered

Marked out of 1.00

Flag question

Select a statement that is INCORRECT about virtual memory.

- Virtual memory and main memory are divided equal sized pages.
- The entire address space required by a process need to be in main memory at once.
- Pages allocation to a process do not need to be stored contiguously.
- Virtual memory uses internal disk storage like hard disk.

Question 18

Not yet answered

Marked out of 1.00

Flag question

A system uses a control memory of 2048 words of 32 bits each. The microinstruction has 3 fields and a 16-bit micro-operations. How many bits are there in the selection field ?

- 5 bits
- 5 kb
- 5 bytes
- 6 bits

Question 19

Not yet answered

Marked out of 1.00

Flag question

Which of the following memory having access times of '3ns - 10ns' to the processor?

Select one:

- Magnetic Tape
- Level 1 Cache
- Main Memory
- Optical Disk

Time left 0:10:20

Question 21

Not yet answered

Marked out of 1.00

Flag question

Which ones from below are the input-output control methods?

I. Channel

II. Interrupt-driven

III. Programmed

IV. Direct Memory Active

- I, II and III
- II and IV
- I, II, III and IV
- I and III

Question 20

Not yet answered

Marked out of 1.00

Flag question

Average memory access time (AMAT) is computed using the following parameters EXCEPT:

- Hit time
- Miss rate
- Miss penalty
- Penalty hit

Question 22

Not yet answered

Marked out of 1.00

Flag question

A common measure of performance for a processor is the rate at which instructions are executed and expressed as _____.

- IPC
- CPI
- FLOPS
- MIPS

Question 23

Not yet answered

Marked out of 1.00

Flag question

How does control bus assist the process execution on the processor?

- Determine the location of the destination
- Determine the location of the source
- Determine when each device can access the bus
- Convey bits from one device to another

Question 24

Not yet answered

Marked out of 1.00

Flag question

The Least Frequently Used (LFU) Replacement policy can be used in which of the following scheme?

- Direct Mapping.
- Direct Mapping and Set Associative Mapping.
- Direct Mapping and Fully Associative Mapping.
- Fully Associative Mapping.

Time left 0:06:41

Question 26

Not yet answered

Marked out of 1.00

Flag question

All instructions required a constant value of _____ to achieve the same number of clock cycles for a processor.

- MIPS
- FLOPS
- CPI
- Hz

Question 27

Not yet answered

Marked out of 1.00

Flag question

Determine why disk drives can be randomly accessed?

- Disk tracks are consecutively numbered starting with track 0
- Error-correcting code in intertrack gap allows random access
- The read/write heads mounted on a comb can swings directly to read the disk
- Sectors on the disk have their own unique address that can be accessed independently

Question 25

Not yet answered

Marked out of 1.00

Flag question

Which of the following is/are basic form/forms of locality:

i - Temporal locality

ii - Spatial locality

iii - Equal locality

iv - Parallel locality

Select one:

- iv only
- i and ii
- ii and iv
- i, ii and iii

Question 28

Not yet answered

Marked out of 1.00

Flag question

What are the 2 principal parts of the CPU?

- Control unit and ALU
- Data path and ALU
- ALU and register
- Data path and control unit

Question 29

Not yet answered

Marked out of 1.00

Flag question

Select a statement that is INCORRECT about write through policy.

- Write through creates a lot of traffic.
- Write through writes to main memory and cache.
- Write through slows down the write process.
- Write through will initially updates data in cache.

Time left 0:03:55

Question 31

Not yet answered

Marked out of 1.00

Flag question

What is the basic functional elements of Internal data paths

- Link register to memory and I/O modules
- Used to store data (also status) internal to the processor
- Used to move data between registers and between register and ALU
- The functional essence of the computer

Question 32

Not yet answered

Marked out of 1.00

Flag question

Which statements are TRUE about RAID 6 ?

I. It supports dual redundancy

II. Data can still be recovered after three disks fail simultaneously

III. Two block parity striping calculated across all disks

IV. Lower performance than RAID 5 for write operation

- I, III and IV
- I, II, III and IV
- I and III
- II and IV

Question 30

Not yet answered

Marked out of 1.00

Flag question

Select the INCORRECT statement:

- Cache memory is slower than main.
- The principle of spatial locality says that if a program accesses one memory address, there is a good chance that it will also access other nearby addresses.
- Cache memory is a small, fast and expensive memory.
- A cache miss occurs if the cache does not contain the requested data

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Time left 1:29:33

Question 1

Not yet answered

Marked out of 11.00

Flag question

1. Refer to the table and figure below, and answer the following questions.

Table 1

3001	Mov al, [3003]
3002	00
3003	76
aaa	bbb
6000	24
6001	00D

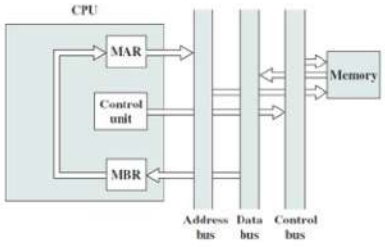


Figure 1: Data Flow

a) Which of the following instruction cycles is not involved in this data flow?

b) During the initial execution, what is the address that will be copied into the MAR?

Time left 1:29:07

a) Which of the following instruction cycles is not involved in this data flow?

b) During the initial execution, what is the address that will be copied into the MAR?

c) Next, what is the instruction to describe the data from address in Q(a) is transferred to the Data bus.

d) What is the value of the PC once the data is transferred to the Data bus?

e) Assume the data flow is an indirect cycle. Once the PC is incremented by 1, what is the address that will be copied from IR into MAR?

f) What is the content that will be stored in the destination register?

g) Choose a task that will not involve in the execute cycle?

2. Solve for the 8 bit arithmetic operation of *2Bh-85h* and choose the value of the corresponding flags.

a) Solve 2Bh-85h ***Note: Answer with Caps Lock and NO white space (i.e: 0A000000)* = h

b) Carry flag =

c) Sign flag =

d) Auxiliary flag =

e) Parity flag =

f) Zero flag =

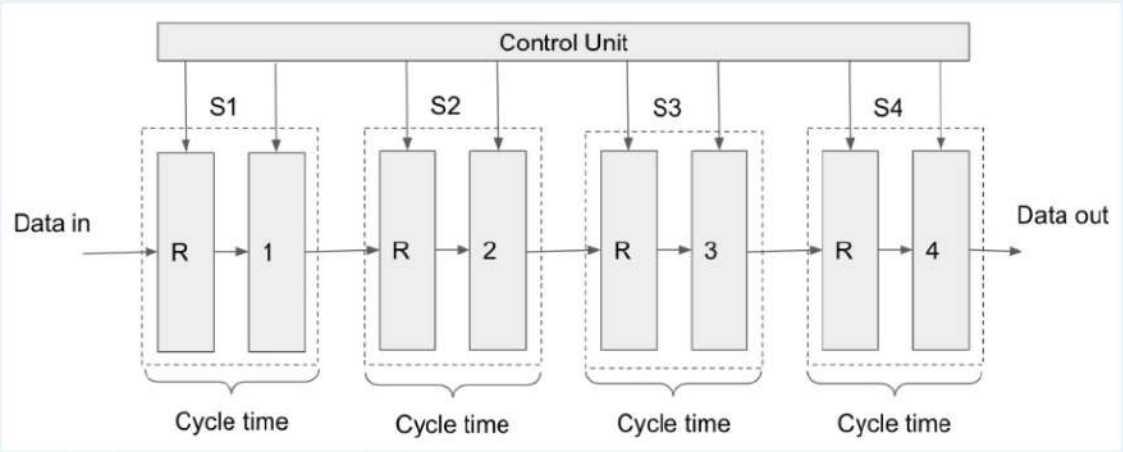
Time left 1:21:56

Question 2

Not yet answered

Marked out of 12.00

Flag question



1. Figure above shows 4 segments of pipeline whereby each has a delay time as follow :

- S1 : 40 ns
- S2 : x ns
- S3 : 2x ns
- S4 : 5+2x ns

The delay time for each of the interface registers are 5 ns.

a) Calculate the delay time, x for S2 if the execution of 100 tasks using non pipeline is 7500 ns

The answer is [ ] ns

g) Overflow flag =

- S4 : 5+2x ns

The delay time for each of the interface registers are 5 ns.

a) Calculate the delay time, x for S2 if the execution of 100 tasks using non pipeline is 7500 ns

The answer is [ ] ns

b) Calculate the total cycle time for non pipeline and pipeline

The answer for non-pipeline is [ ] ns and pipeline is [ ] ns

c) Calculate the execution time for 150 tasks for non pipeline and pipeline

The answer for non-pipeline is [ ] ns and pipeline is [ ] ns

2. Consider 4 segments in a pipeline with branching problem. Based on the following instruction set, answer the question below

```
I1:     MOV    AX, BX
I2:     ADD    BX, CX
I3:     INC    CX
I4:     JMP    DDK
I5:     MOV    AX, 1
I6:     INC    AX
...
I12:DDK    DEC    BX
```

a) How many NOP needed

The answer is [ ]

Time left 1:20:00

a) How many NOP needed

The answer is

b) Rewrite the new instruction set by including NOP

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Time left 0:53:02

Question 4
Incomplete answer
Marked out of 10.00
Flag question

QUESTION (10 marks)

Consider a sample of five programs executes on 3 different systems as shown in Table 1.

If System B and System C are normalized to System A, answer the following questions.

(Note: All answers must in 4 decimal point numbers. Example 0.1234, 1.0000)

Table 1: Execution time for systems A, B, and C

Program	System A	System B	System C
v	75	200	150
w	200	250	300
x	250	175	200
y	400	800	500
z	1500	1200	1000

(a) Complete Table 2 by filling up the normalized execution time for each program in each system. [7 marks]

Table 2: Normalized execution time for systems A, B, and C

Program	System A	System B	System C
v			
w			0.8333
x	0.7000		
y		1.000	
z			1.2000

(b) Calculate the Geometric Mean for all systems. [3 marks]

	System A	System B	System C
Geometric Mean			

Time left 0:58:03

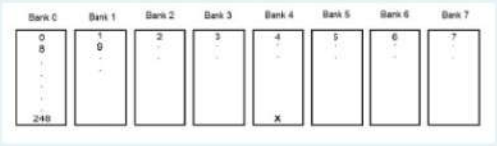
Question 3

Not yet answered

Marked out of 9.00

Flag question

Figure illustrate the memory bank of a computer system. Same banks indicate the locations or address which in sequential order. The size is 1 byte each. Answer the following questions. ANSWER WITH NO WHITE SPACE (EXAMPLE : 101010101101)



a) What is the memory interleaving approach used? =

b) How many bits are used to represent the bank? = bits

c) What is the total memory capacity of each bank? = bits

d) How many bits are used to represent each address? = bits

e) Complete the table by writing the bank number for each address.

Address (Hex)	Bank
15h	
96h	
DBh	
7Ch	

Time left 0:47:34

Question 5

Not yet answered

Marked out of 12.00

Flag question

Given that a 4-way set associative cache memory has 64 KB data and each block contains 32 bytes. The main memory capacity is 4 GB.

a. Find the number of bits for the main memory address. ANSWER: [ ] bits

b. How many blocks are there in a set? ANSWER: [ ] blocks

c. How many sets the cache has? ANSWER: [ ] sets

d. The main memory address format is => | Tag: [ ] bits | Set: [ ] bits | Word: [ ] bits |

e. Which set will be mapped by the main memory address 458195h. ANSWER: [ ] (in decimal)

Question 1

Not yet answered

Marked out of 1.00

Flag question

Which of the following is NOT the advantage of High-Order Interleaving (HOI)?

Select one:

- Use with or without conflict problem in multiprocessors if the modules are partitioned
- Provide better reliability
- Produces memory interference
- Provide memory extension

Question 2

Not yet answered

Marked out of 1.00

Flag question

All micro-operations fall into the following categories, which statement is NOT TRUE

- Perform an arithmetic or logic operation using registers for input and output
- Transfer data from one register to another register
- Transfer data from an external interface to a register
- Link registers to memory and I/O modules, often by system bus

Question 5

Not yet answered

Marked out of 1.00

Flag question

What are the differences between Blu-ray DVDs and HD-DVDs?

I. Blu-ray DVDs have greater storage capacity compared to HD-DVDs

II. HD-DVDs are backward compatible with red laser DVDs but Blu-ray DVDs are not

III. Blu-ray DVDs are backward compatible with red laser DVDs but HD-DVDs are not

IV. Blu-ray DVDs are not designed to store high-definition videos like HD-DVDs

- ○ I and III
- ○ III and IV
- ○ I and II
- ○ II, III and IV

Question 3

Not yet answered

Marked out of 1.00

Flag question

What will happen to the processor if the interrupt is disabled?

- The processor will schedule the next execution of interrupt
- The processor will execute the request signal
- The processor will ignore the interrupt request signal
- The processor will delete the request signal

Question 4

Not yet answered

Marked out of 2.00

Flag question

A program runs in 20 seconds on a computer, which has an 8 MHz clock. Calculate the CPU clock cycle for the computer.

- 4×10^5 cycles
- 160×10^6 cycles
- 2.5×10^6 cycles
- 1.25×10^5 cycles

Question 6

Not yet answered

Marked out of 1.00

Flag question

Which register stores the word most recently read?

- Program Counter
- Memory Address Register
- Instruction Register
- Memory Buffer Register

Question 7

Not yet answered

Marked out of 1.00

Flag question

Control Unit design can be organized systematically and avoided complexity using _____ approaches.

i. Hardwired
ii. Microprogrammed
iii. Parallel
iv. Microinstruction

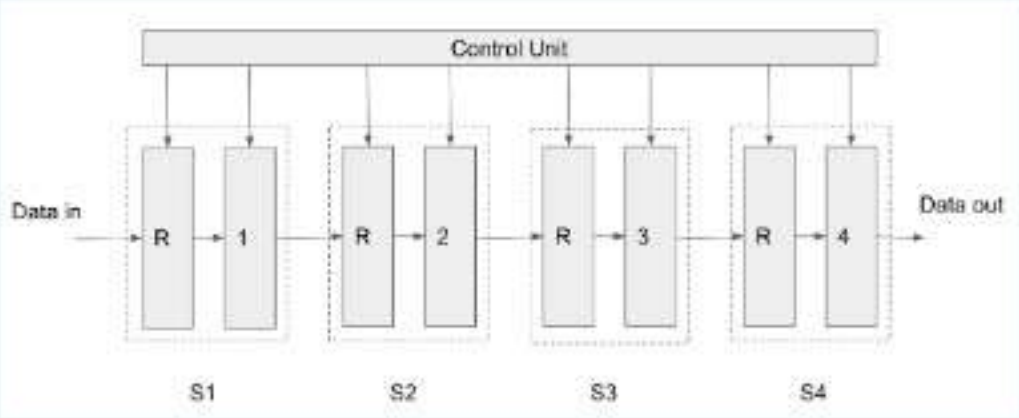
- iii and iv
- ii and iii
- i and ii
- i only

Question 9

Not yet answered

Marked out of 1.00

Flag question



A 4-stage pipeline has stage delays as 150, 120, 150 and 140 ns. Registers were used between stages and had a delay of 5 ns each. Assuming constant clock rate, what is the total time taken to process 1000 data items on this pipeline.

- 165495 ns
- 155465 ns
- 160500 ns
- 169555 ns